

EME 150B – Mechanical Design

Homework assignment 3

(Please give the full equations for full points.)

A precision, enclosed gear drive consists of a 20° spur pinion having 16 teeth driving a 52-tooth gear. The power of 4 hp is to be transmitted. The pinion speed is 300 rev/min, the face width 2 in, and the diametral pitch 5 teeth/in. The gears are grade 1 steel, through-hardened at 200 Brinell, made to No. 6 quality standards ($Q_v = 6$), uncrowned, and are to be accurately and rigidly mounted. Assume a pinion life of 10^8 cycles and a reliability of 0.90. Assume uniform loading, $K_o = 1$, and constant thickness of the gears ($K_B = 1$). Use AGMA safety factors S_F and $S_H = 1$.

Use the appropriate figures, tables and information in Shigley's Chapter 14. Make sure to state your assumptions.

a) Determine the AGMA bending stress, σ_b , for the pinion and the bending fatigue factor of safety, n_b .

Solution range: The bending stress is between 6,000 – 7,500 psi, the factor of safety is between 4.0 – 5.5

b) Determine the AGMA contact stress, σ_c , for the pinion and the contact-stress fatigue factor of safety, n_c .

Solution: The contact stress is between 65,000 – 75,000 psi, the factor of safety is between 1 – 2

c) Comment on your factors of safety. Which one would you try to improve and how?

d) Answer based on your reading of **Chapter 14-1** and engineering expertise: Where do you expect the highest loads occur on gear teeth in practical applications? Would it be in the middle of the tooth or the tip?